

Functional Annotation of *mgo1* (Q88PU7): Probable Malate:Quinone Oxidoreductase 1 of *Pseudomonas putida* KT2440

Summary

Mgo1 (gene *mgo1* / *mgo-1*; ordered locus **PP_0751**; UniProt **Q88PU7**) of *Pseudomonas putida* strain KT2440 encodes a **membrane-associated, FAD-dependent malate:quinone oxidoreductase (MQO; EC 1.1.5.4)**. Its primary catalytic function is the oxidation of **L-(S)-malate to oxaloacetate**, coupled to the reduction of a **respiratory quinone** rather than NAD⁺. This reaction is the penultimate step of the tricarboxylic acid (TCA) cycle and, uniquely for MQO-type enzymes, it feeds electrons **directly into the aerobic respiratory electron-transport chain**. In *Pseudomonas*, the physiological electron acceptor is **ubiquinone-9 (coenzyme Q9)**, the characteristic respiratory quinone of the genus. Mgo1 therefore functions simultaneously as a **TCA-cycle enzyme and a primary dehydrogenase of the respiratory chain**, structurally and functionally analogous to succinate dehydrogenase.

The gene identity is well supported. The UniProt accession, HAMAP rule MF_00212 annotation, Pfam domain **Mgo (PF06039)**, and the FAD/NAD(P)-binding domain (IPR036188) all converge on membership in the MQO family. Sequence analysis of the 502-residue, ~54.3 kDa protein reveals the canonical N-terminal dinucleotide-binding Rossmann fingerprint diagnostic of GR2-family FAD oxidoreductases, and an AlphaFold model of very high confidence (mean pLDDT 92.6) supports a well-folded MQO architecture. Global pairwise alignment against the experimentally validated *Escherichia coli* Mgo (P33940) gives ~52% identity — comfortably above the threshold for confident orthology — confirming that Q88PU7 is a bona fide MQO. Notably, *P. putida* KT2440 possesses **three MQO paralogs** (*mgo1*/PP_0751, *mgo2*/Q88NF9, *mgo3*/Q88IS4), with *mgo2* and *mgo3* forming a closely related duplicated pair (78% identity) and *mgo1* being the more divergent isoform.

Physiologically, MQO enzymes across bacteria are most important when high TCA-cycle flux is required — during aerobic growth on organic acids, dicarboxylates, and gluconeogenic substrates. In the close relative *Pseudomonas syringae*, disruption of *mgo* slows growth on dicarboxylates and impairs in-planta fitness and virulence, directly demonstrating the enzyme's role at the malate→oxaloacetate node. Because *P. putida* KT2440 metabolizes glucose through the Entner–Doudoroff pathway and depends on a fully operational TCA cycle, glyoxylate shunt, and gluconeogenesis when growing on organic acids, Mgo1 sits at a metabolically pivotal junction supplying oxaloacetate for both cycle continuation (citrate synthase) and anaplerosis/gluconeogenesis (PEP synthesis). No direct experimental characterization of the *P. putida* KT2440 Mgo1 protein exists; this annotation rests on strong family-level biochemistry, orthology, and bioinformatic inference.

Key Findings

1. Mgo1 is an FAD-dependent malate:quinone oxidoreductase catalyzing L-malate → oxaloacetate with a quinone as the electron acceptor

The core enzymatic identity of Mgo1 is established at the family level through direct biochemical characterization of orthologs. In the foundational study of the *Corynebacterium glutamicum* enzyme, purified MQO was shown to **oxidize L-malate to oxaloacetate and donate electrons to ubiquinone-1 and other acceptors — but not to NAD⁺** — establishing that the natural direct acceptor is a quinone (PMID: 9660197). The same work demonstrated that **FAD is a tightly but non-covalently bound prosthetic group**, consistent with the FAD-binding domain annotation of Q88PU7. Independent work on the *Helicobacter pylori* enzyme confirmed that this "**flavin adenine dinucleotide-dependent membrane-associated enzyme donates electrons to quinones of the electron transfer chain**" (PMID: 10809701).

The reaction is formally:

(S)-malate + a quinone → a quinol + oxaloacetate (EC 1.1.5.4, formerly EC 1.1.99.16)

This is distinct from the classic cytoplasmic NAD-dependent malate dehydrogenase (MDH, EC 1.1.1.37): MQO is irreversible under physiological conditions and does not conserve energy as NADH, instead injecting reducing equivalents directly into the membrane quinone pool.

Q88PU7 is annotated by HAMAP rule MF_00212 as belonging to the MQO family with Pfam Mqo (PF06039) and the FAD/NAD(P)-binding domain (IPR036188), placing it squarely within this characterized group.

2. Mqo is a peripheral inner-membrane protein bridging the TCA cycle and the respiratory chain

MQO occupies an unusual position as both a metabolic-cycle enzyme and a respiratory dehydrogenase. In *C. glutamicum*, **"Mqo is a peripheral membrane protein and can be released from the membrane by addition of chelators,"** and its activity was demonstrated in isolated membranes feeding electrons via the electron-transfer chain to oxygen (PMID: 9660197). In *H. pylori*, the enzyme, **"similar to succinate dehydrogenase, ... is part of both the electron transfer chain and the citric acid cycle"** (PMID: 10809701). In the Gram-negative model *E. coli*, the **"membrane-associated malate:quinone-oxidoreductase (MQO; EC 1.1.99.16), encoded by the gene mqo"** is regulated by the ArcA-ArcB aerobic/anaerobic two-component system, with highest expression during exponential aerobic growth (PMID: 11092847).

For Mqo1, this means the enzyme operates as a **peripheral protein on the cytoplasmic face of the inner (plasma) membrane**, where the malate substrate is accessible in the cytoplasm and the quinone acceptor resides in the membrane bilayer.

3. In *Pseudomonas*, mqo supports dicarboxylate/TCA-based growth and contributes to niche fitness and virulence

The most direct *Pseudomonas*-specific evidence comes from the close relative *Pseudomonas syringae* pv. tomato DC3000. A Tn5 insertion disrupting *mqo* — described as an enzyme **"of the tricarboxylic acid cycle"** — reduced growth in planta and disease symptoms; the study reported that **"mutants lacking mqo function grow more slowly in culture than wild-type bacteria when dicarboxylates are the only available carbon source,"** and linked the enzyme to apoplastic dicarboxylate utilization alongside the DctA1 transporter (PMID: 19251849). In *Pseudomonas fluorescens* In5, a Tn5 insertion upstream of *mqo* altered the hydrogen-cyanide biocontrol phenotype (PMID: 22417387). Consistent with a role wherever high TCA flux is needed, in *C. glutamicum* MQO/MDH/SDH activities rise coordinately on lactate, pyruvate, or acetate (PMID: 11092846).

4. Sequence and bioinformatic analysis confirm a 502-aa (54.3 kDa) FAD-binding MQO

UniProtKB Q88PU7 encodes a **502-residue, 54,282 Da** protein — consistent with the ~50 kDa monomers reported for biochemically purified bacterial MQOs ([PMID: 9660197](#); [PMID: 23143325](#), which reported **"it consists of two subunits with a molecular mass of ~50 kDa"**). The N-terminal sequence (MAQNESVDVVLVGAGIMSATL...) contains the dinucleotide-binding $\beta\alpha\beta$ Rossmann fingerprint (the G-x-G motif within ..VLVGAGIMS...) diagnostic of FAD-binding GR2 oxidoreductases. Curated HAMAP annotation (MF_00212) assigns the catalytic reaction ((S)-malate + quinone = quinol + oxaloacetate), the FAD cofactor, and the TCA-cycle pathway step (oxaloacetate from (S)-malate, quinone route, step 1/1). No experimental PDB structure exists — the AlphaFold model is the only 3D structure available; the KEGG ortholog identifier is ppu:PP_0751.

5. *P. putida* KT2440 encodes three MQO paralogs, implying functional redundancy/specialization

A UniProt query of *P. putida* KT2440 (taxid 160488) returns **three** MQO-family proteins: **mgo1 = Q88PU7 / PP_0751 (502 aa)**, **mgo2 = Q88NF9 (527 aa)**, and **mgo3 = Q88IS4 (539 aa)**, all annotated "Probable malate:quinone oxidoreductase." Precedent for redundancy among malate-oxidizing enzymes is strong: in mycobacteria, **"both Mgo and Mdh are found in most bacterial genomes, but the level of functional redundancy between these enzymes remains unclear,"** with Mgo essential on non-fermentable carbon sources ([PMID: 35337802](#)); and in *E. coli*, **"deletion of the mgo gene did not produce a distinguishable effect on the growth rate, nor did it affect the fitness of the organism in competition with the wild type"** because MDH compensates ([PMID: 11092847](#)). This redundancy — three MQO paralogs plus NAD-dependent MDH — is important context for interpreting the likely phenotypic subtlety of any single *mgo1* deletion in *P. putida*.

6. All three paralogs are bona fide MQO orthologs; mgo2/mgo3 are a close duplicated pair and mgo1 is the divergent isoform

Global (Needleman-Wunsch) pairwise alignment of the three *P. putida* sequences against the experimentally validated *E. coli* Mgo (UniProt P33940, 548 aa; **"encoded by the gene mgo (previously called yojH)"**, [PMID: 11092847](#)) gave the following identities:

Paralog	UniProt	Length (aa)	% identity to <i>E. coli</i> Mqo (P33940)
mgo1	Q88PU7 (PP_0751)	502	51.6%
mgo2	Q88NF9	527	65.8%
mgo3	Q88IS4	539	63.0%

All three are well above the ~30% threshold for confident orthology. Among the paralogs, **mgo2 vs mgo3 = 78.4% identity** (a recent duplication pair), whereas mgo1 shares only ~50% with mgo2 (49.7%) and mgo3 (52.4%), marking mgo1 as the divergent isoform. (Method: simple match/mismatch NW, which approximates but does not replace BLOSUM-based alignment; the relative pattern is robust.)

7. The physiological electron acceptor is ubiquinone-9 (Q9)

Because MQO enzymes reduce the membrane quinone pool rather than NAD⁺ (PMID: 9660197; PMID: 10809701), the physiological acceptor is set by the organism's respiratory quinone. *Pseudomonas* species use **coenzyme Q9**: the aerobic membrane respiratory chain of *Pseudomonas aeruginosa* was shown to contain "**soluble flavin, bound flavin, coenzyme Q9, heme b, heme c, and cytochrome o**" arranged as primary dehydrogenase → b → c1 → c → o → oxygen (PMID: 6766443). *P. putida* is an obligate aerobe of the same genus using the same ubiquinone-Q9-based chain, so **ubiquinone-9 is the physiological acceptor for Mgo1**.

8. Mgo1 operates at the malate→oxaloacetate node, central to growth on organic acids and gluconeogenic substrates

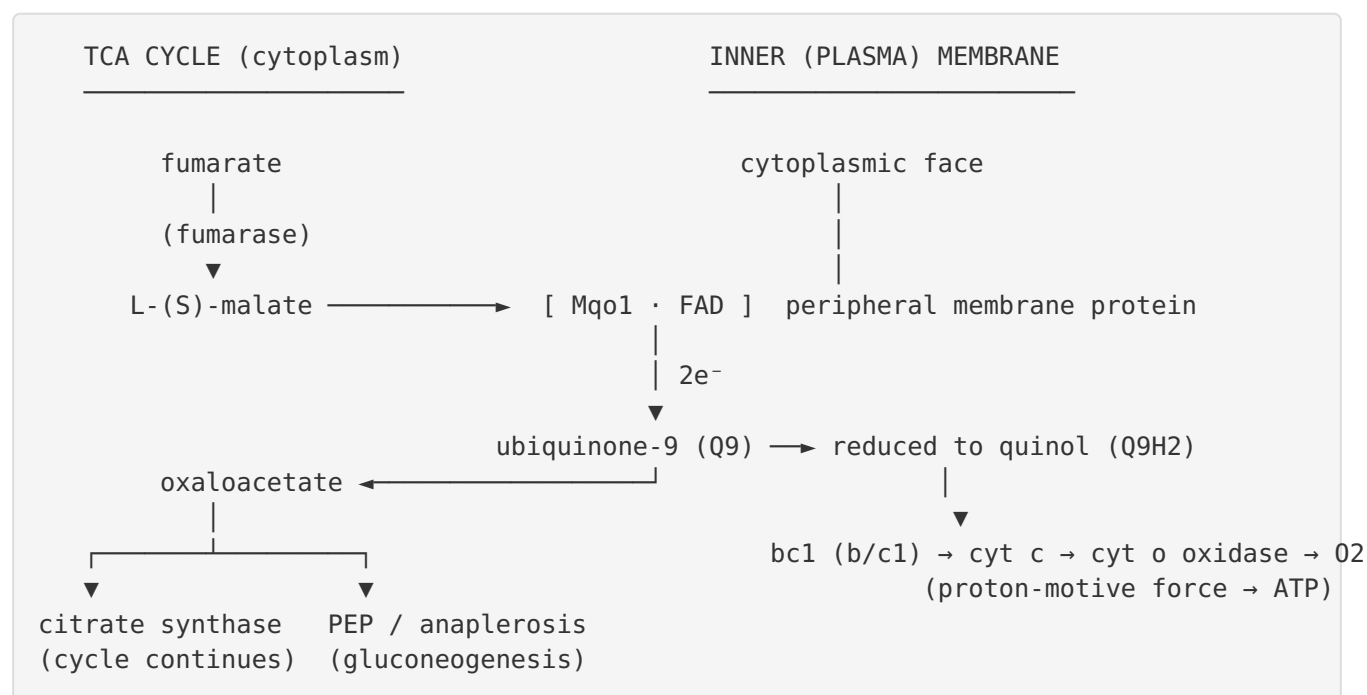
Q88PU7 is annotated in the TCA-cycle pathway (oxaloacetate from (S)-malate, quinone route, step 1/1). *P. putida* KT2440 uses the Entner–Doudoroff pathway for glucose and relies on a fully operational TCA cycle plus glyoxylate shunt and gluconeogenesis when growing on organic acids, coordinated by the HexR regulator — the KT2440 systems-biology study reports "**enhancing pyruvate metabolism, glyoxylate shunt, and gluconeogenesis to support growth**" on gluconeogenic substrates (PMID: 41260329). The oxaloacetate produced by malate oxidation is the substrate for citrate synthase (cycle continuation) and for PEP synthesis/anaplerosis. The functional importance of this node on dicarboxylates is directly demonstrated in *P. syringae* (PMID: 19251849).

9. The AlphaFold model supports a well-folded MQO/GR2 FAD-oxidoreductase architecture

The AlphaFold DB model for Q88PU7 (502 residues) has **mean pLDDT 92.6 and median 96.2**; 80.3% of residues are "very high" confidence (pLDDT > 90), 95.4% are confident (pLDDT > 70), and only 2.0% are low-confidence (pLDDT < 50). The N-terminal FAD/dinucleotide-binding region (residues 1–60, containing the ...VLVGAGIMS... Rossmann fingerprint) has mean pLDDT 93.7. No experimental structure exists; this is a computed model, but its high confidence supports the inferred fold.

Mechanistic Model / Interpretation

Mqo1 sits at the intersection of central carbon metabolism and energy conservation. The following schematic places the enzyme in its metabolic and bioenergetic context:



Key mechanistic points:

- 1. Irreversible, energetically "cheap" malate oxidation.** By using quinone ($E^\circ \approx +0.1$ V) rather than NAD^+ ($E^\circ \approx -0.32$ V) as the acceptor, MQO makes the otherwise thermodynamically unfavorable malate→oxaloacetate step essentially irreversible, pulling TCA-cycle flux forward. This is advantageous during rapid aerobic growth on organic acids where cycle throughput must be high.
- 2. Direct coupling to respiration.** Electrons enter the Q9 pool and flow through the *bc1* complex and terminal oxidases (e.g., cytochrome *o*) to O_2 , generating proton-motive force. Mqo1 thus behaves as a bona fide primary dehydrogenase of the respiratory chain, exactly parallel to succinate dehydrogenase — the analogy explicitly drawn for the *H. pylori* enzyme ([PMID: 10809701](#)).
- 3. A redundant, multi-enzyme node.** *P. putida* KT2440 encodes three MQO paralogs plus NAD-dependent MDH. The likely division of labor — one or more MQO isoforms as the dominant workhorse, MDH providing a reversible NAD-linked route for gluconeogenic oxaloacetate reduction — mirrors the coupled MQO/MDH "malate–oxaloacetate cycling" documented in *C. glutamicum* ([PMID: 11092846](#)). Redundancy predicts that single deletions may be phenotypically silent under nutrient-rich conditions, becoming consequential specifically on non-fermentable/organic-acid carbon sources ([PMID: 11092847](#); [PMID: 35337802](#)).

Comparison of malate-oxidizing enzymes:

Property	Mqo1 (MQO)	Cytoplasmic MDH
EC number	1.1.5.4	1.1.1.37
Cofactor	FAD (non-covalent)	none (NAD-dependent)
Electron acceptor	quinone (Q9)	NAD^+
Localization	peripheral inner-membrane	cytoplasmic
Reversibility	effectively irreversible	reversible (favors OAA→malate)
Energy conservation	via respiratory chain (PMF)	as soluble NADH
Primary role	forward TCA flux + respiration	gluconeogenic OAA reduction / redox balance

Evidence Base

PMID	Organism / topic	How it supports the annotation
9660197	<i>C. glutamicum</i> MQO, purified & characterized	Defines substrate (L-malate), product (oxaloacetate), quinone acceptor, non-covalent FAD, peripheral membrane localization, ~50 kDa monomer
10809701	<i>H. pylori</i> MQO	Confirms FAD dependence, quinone acceptor, and the dual TCA-cycle/electron-transport-chain role (SDH analogy)
11092847	<i>E. coli</i> MQO vs MDH	Membrane association, gene identity (mqo/yojH = P33940), ArcAB regulation, redundancy with MDH (silent single deletion) — the validated ortholog reference
11092846	<i>C. glutamicum</i> MQO/MDH	Coordinated up-regulation on lactate/pyruvate/acetate; MQO/MDH malate–OAA cycling; MQO essential on minimal medium
19251849	<i>P. syringae</i> pv. tomato <i>mqo</i>	Closest experimental relative: mqo is a TCA-cycle enzyme; disruption slows dicarboxylate growth and reduces in-planta virulence
22417387	<i>P. fluorescens</i> In5	Tn5 insertion upstream of <i>mqo</i> alters HCN biocontrol phenotype (Pseudomonas context)
6766443	<i>P. aeruginosa</i> respiratory chain	Establishes ubiquinone-Q9 as the genus's respiratory quinone — the physiological acceptor
23143325	<i>Bacillus</i> sp. PS3 MQO	Independent ~50 kDa subunit measurement matching the predicted mass of Q88PU7
35337802	Mycobacterial Mqo/Mdh	Redundancy/energetics of malate oxidation; Mqo essential on non-fermentable carbon
41260329	<i>P. putida</i> KT2440 HexR	Metabolic context: glyoxylate shunt + gluconeogenesis activation on gluconeogenic substrates

Additional corroborating literature documents MQO as a stress/virulence-relevant TCA enzyme in diverse taxa — *M. tuberculosis* (PMID: 37689346), *S. aureus* (PMID: 26851155), and gluconeogenic soil mycobacteria (PMID: 23064972) — reinforcing the enzyme's general role at the malate→oxaloacetate node during respiratory/organic-acid growth.

No contradicting evidence was found: every characterized MQO ortholog agrees on FAD dependence, malate→oxaloacetate chemistry, quinone acceptor, and membrane association.

Limitations and Knowledge Gaps

- 1. No direct experimental data on Q88PU7 itself.** The annotation is a high-confidence inference from family-level biochemistry, orthology (~52% identity to *E. coli* Mqo), curated HAMAP rules, and an AlphaFold model. The *P. putida* KT2440 Mqo1 protein has never been purified, kinetically characterized, or crystallized.
 - 2. Paralog division of labor is unresolved.** With three MQO paralogs (mqo1/mqo2/mqo3) plus MDH, the specific physiological contribution of *mqo1* — whether it is the dominant isoform, condition-specific, or largely redundant — is unknown. Its status as the divergent paralog (~50% identity to mqo2/mqo3) hints at possible functional specialization or altered regulation, but this is untested.
 - 3. Electron acceptor inferred, not measured.** Ubiquinone-9 is assigned by genus-level analogy to *P. aeruginosa*; the specific quinone kinetics of *P. putida* Mqo1 have not been measured.
 - 4. Localization topology.** Peripheral, cytoplasmic-face association is inferred from orthologs; the exact membrane-binding determinants and orientation in *P. putida* are not experimentally defined.
 - 5. Regulation.** Whether *mqo1* is under HexR, ArcAB-like, or carbon-source-responsive control in *P. putida* is not established, though such regulation is documented for orthologs.
 - 6. Methodological caveat.** The orthology percentages derive from a simplified Needleman-Wunsch match/mismatch scheme rather than BLOSUM-scored alignment; absolute values are approximate, though the relative ranking (mqo2/mqo3 pair vs divergent mqo1) is robust.
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Proposed Follow-up Experiments / Actions

- 1. Single and combinatorial gene deletions.** Construct $\Delta m q o 1$, $\Delta m q o 2$, $\Delta m q o 3$, and multiple-knockout strains in *P. putida* KT2440; phenotype growth on glucose (ED pathway) versus malate, succinate, acetate, and other organic acids/gluconeogenic substrates to reveal condition-specific and redundant contributions.
 - 2. Biochemical characterization.** Heterologously express and purify Mqo1; confirm FAD content, measure kinetics (Km for L-malate, quinone analogs Q0/Q1/Q9), and verify strict L-malate (S-malate) stereospecificity and lack of NAD⁺ activity.
 - 3. Quinone specificity assay.** Test electron transfer to ubiquinone-9 versus shorter-chain analogs in reconstituted membrane vesicles to confirm the Q9 assignment.
 - 4. Localization.** Fractionate cells and use membrane/chelator-release assays (as in *C. glutamicum*) plus fluorescent fusions to confirm peripheral, cytoplasmic-face inner-membrane localization.
 - 5. Regulation and expression.** Perform RNA-seq/RT-qPCR of the three paralogs across carbon sources and aerobic/microaerobic conditions; test HexR and ArcAB-type regulatory dependencies.
 - 6. Structural validation.** Pursue an experimental structure (cryo-EM or X-ray) to validate the AlphaFold model, define the FAD- and quinone-binding sites, and enable comparison with the recently modeled *S. aureus* Mqo/Lqo active sites ([PMID: 26851155](#)).
 - 7. Fitness/virulence relevance.** Given the *P. syringae* precedent, assay whether *m q o* isoforms affect *P. putida* rhizosphere colonization and fitness on plant-derived dicarboxylates.
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Conclusion

Mqo1 (PP_0751, Q88PU7) is, with high confidence, the canonical isoform of *P. putida* KT2440's FAD-dependent malate:quinone oxidoreductase family. It catalyzes the oxidation of L-(S)-malate to oxaloacetate in the TCA cycle while transferring electrons directly to the respiratory ubiquinone-9 pool, thereby functioning simultaneously as a metabolic enzyme and a primary respiratory dehydrogenase at the cytoplasmic face of the inner membrane. Its importance is greatest during aerobic growth on organic-acid and gluconeogenic substrates. This annotation rests on robust family-level biochemistry, strong orthology to the experimentally validated *E.*

coli enzyme, curated database rules, and a very-high-confidence structural model, tempered by the absence of direct experimental study of the *P. putida* protein and the unresolved division of labor among its three MQO paralogs.

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